MILITARY SPECIFICATION

POWER SUPPLY, TRANSISTORIZED

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of transistorized power supply unit for use with DY-84/ARN-14A.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-E-5400	Electronic Equipment, Airborne, General Specification For
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-E-17555	Electronic And Electrical Equipment, Accessories, And Repair Parts; Packaging And Packing Of
MIL-S-19500	Semiconductor Devices, General Specification For

STANDARDS

Military

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking Of US Military Property
MIL-STD-454	Standard General Requirements For Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements For Equipment
MIL-STD-701	Lists Of Standard Semiconductor Devices
MIL-STD-831	Test Reports, Preparation Of

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DRAWINGS

Air Force

61C34406 Transistorized Power Converter Unit

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction testing. This specification makes provision for preproduction testing.

3.2 General specification. The requirements of MIL-E-5400 apply as requirements to this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.2.1 Exceptions to the requirements of MIL-E-5400.

3.2.1.1 Reordered production equipment. For this specification, "Reordered production equipment" of MIL-E-5400 is amended to the extent that all nonstandard parts shall be submitted for approval of the procuring activity, for each procurement, whether previously approved or not.

3.2.1.2 Time schedule of approved requests. For this specification, "Time schedule of approved requests" of MIL-E-5400 is amended to the extent that the Government shall not incur any liability to compensate the contractor for any disapproved nonstandard parts which are ordered or fabricated prior to receipt of notification.

3.3 Design and construction.

3.3.1 Physical size. The power supply envelope dimensions shall conform to Drawing 61C34406 and shall not exceed 3.5 pounds in weight.

3.3.2 Design. The power supply shall be, electrically and mechanically, a direct interchangeable replacement for dynamotor (DA-183) as used with the DY-84/ARN 14A. The power supply shall operate as a Direct Current (DC) to DC power supply rated for continuous duty.

3.3.3 Test connections. All electrical tests shall be made with the power supply connected to the DY-84/ARN-14A and all output measurements taken at the output terminals of the DY-84/ARN-14A.

3.3.4 Warning markings. All power supply units shall have warning markings, as follows: "Do not use with Receivers R-539/ARN-14 or R-539E/ARN-14" and "Direct replacement for DA-183/DY-84/ARN-14A."

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3.3.5 Lead wires. Lead wires shall be 4.5 ±0.5 inches in length. Lead wires shall bear a designation, printed directly on the wire or a nonmetallic sleeve, in a manner as to be clearly visible, and shall have color coded terminals as follows:

Lead	Printed Marking	Terminal Color
Input Positive	27.5 DC IN	Blue
Output Positive	DC OUT	Red
Ground	GND	Black

3.3.6 Transistors. Transistors shall comply with MIL-S-19500 and MIL-STD-701, unless specifically waived by the procuring activity.

3.3.7 Substitution of material. If the manufacturer desires to substitute for a specified or previously approved material or fabricated part, he shall submit to the procuring agency a statement to that effect describing the proposed substitute together with evidence to substantiate his claim that such substitution is adequate. At the discretion of the procuring agency, test samples or additional data may be required to prove the adequacy of the proposed substitute. Use of the substitute will be contingent of the approval of the procuring agency.

3.3.8 Life. The power supply shall deliver rated output for a minimum of 2000 hours and shall have a mean-time-between-failures of at least 500 hours.

3.4 Performance.

3.4.1 Electrical characteristics.

Input Voltage - 27.5 volts DC

Input Current - 3.0 amperes (maximum)

Output Voltage - 270 -8 +22 volts DC

Output Current - 150 Milliamperes (minimum)

Output Ripple Voltage - 1.0 Volt, Peak to Peak

Output Regulation - 25 percent with variable load, 10 percent with variable input

Efficiency - 60 percent (minimum)

3.4.2 Operation at maximum and minimum input. The power supply shall operate within specified limits with no detrimental effect when subjected to input voltages which vary from 22 to 35 volts DC.

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3.4.3 Interference. Interference control requirements shall be in accordance with MIL-STD-461. Special emphasis shall be given to the attenuation of inadvertent radio interference, atmospherics, and switching transients having a time duration of 0.01 to 40 milliseconds.

3.4.4 Overload. The power supply shall operate in a satisfactory manner during and after being subjected to an input of 35 volts DC, surges of 200 volts at the input, arcing short circuits at the output and zero ohms short circuits at the output.

3.4.5 Environmental conditions.

3.4.5.1 Temperature operating limits. The power supply shall be capable of continuous operation over a range of -55° Centigrade (C) to +71°C.

3.4.5.2 Humidity. Upon completion of the specified humidity test of 4.3.3.9, the power supply shall be capable of meeting the electrical requirements of 3.4.1.

3.4.5.3 Altitude. The power supply shall be capable of meeting the requirements for output voltage and current, ripple content, regulation and efficiency at an atmospheric pressure corresponding to an altitude of 70,000 feet.

3.4.5.4 Salt spray. The finishes and coating shall be capable of withstanding a fifty-hour salt spray test of 4.3.3.9 without evidence of corrosion that will be detrimental to the performance of the converter.

3.4.5.5 Vibration. The power supply shall be capable of meeting the output voltage and current, ripple content, efficiency and regulation requirements during and after subjection to the vibration specified herein. There shall be no evidence of electrical or mechanical failure during/after completion of this test.

3.4.5.6 Fungus. Upon completion of the specified fungus test of 4.3.3.9, the power supply shall be capable of meeting electrical requirements of 3.4.1. There shall be no visible growth of fungus on the power supply.

3.4.5.7 Explosion proof. The power supply shall not cause ignition of an explosive vapor surrounding the unit when subjected to the specified explosion test of 4.3.3.9.

3.4.5.8 Shock. Upon completion of the specified shock test of 4.3.3.9, the power supply shall be capable of meeting the requirements for output voltage and current, ripple content, regulation, and efficiency.

3.5 Identification of product. Equipment, assemblies and parts shall be marked for identification in accordance with MIL-STD-130.

3.6 Government loaned property. Upon request, the Government will lend the contractor three each Dynamotor Unit, DY-84/ARN-14A, FSN 6125-635-8795, and three each Mountings, MT-962/ARN-14A, FSN 5826-327-4222.

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3.7 Workmanship. Workmanship shall be in accordance with MIL-STD-454, Requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of tests. The inspection and testing of the converter shall be classified as follows:

a. Preproduction tests.

b. Acceptance tests.

4.3 Preproduction tests. The requirements of MIL-E-5400 apply with the exceptions and additions specified herein.

4.3.1 Test sample. Three first articles shall be tested for design approval. Test specimens may be used by the contractor as a production sample or submitted for acceptance at the end of the contract, provided they are completely refurbished to comply with the requirements stated herein and without increase in the cost of the contract. All parts, specimens, or assemblies destroyed in making tests required by this specification and/or drawings, shall be in addition to the quantity specified in the contract or purchase order and shall be furnished without increase of the cost of the contract or order.

4.3.2 Test report. A certified copy of the contractor's preproduction test data in accordance with MIL-STD-831, shall be submitted to the procuring agency and shall include an analysis of all failures which occurred and contemplated remedial procedures. All inspections and tests, giving the results of tests required to determine compliance with the requirements and tests specified herein, shall be kept complete and shall be available to the Government at all times. The record or Report of Inspection and Test shall be signed and approved by a responsible person specifically designated by the contractor.

4.3.3 Basic operation. Operation of the power supply unit shall be as follows:

4.3.3.1 Output. The output of the power supply shall be determined by measuring and recording the output voltage and current of the unit with rated input applied.

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4.3.3.2 Ripple content. The output ripple content shall be measured with a peak reading voltmeter at rated input.

4.3.3.3 Regulation with variable load. With rated input applied, variable load regulation shall be determined as follows:

$$\frac{100\ (V_1 - V_2)}{V_2} = \%\ \text{regulation}$$

Where: V_1 = Output voltage at 10 milliamperes output current
V_2 = Output voltage at 150 milliamperes output current

Record percent regulation.

4.3.3.4 Regulation with variable input. With 150 milliamperes of output current, measure output voltage after one hour of operation at an input of 22 volts DC, with 150 milliamperes of output current, measure output voltage after one hour operation at 30 volts DC.

$$\frac{100\ (V_1 - V_2)}{V_2} = \%\ \text{regulation}$$

Where: V_1 = Output voltage at 30 volts input
V_2 = Output voltage at 22 volts input

Record percent regulation.

4.3.3.5 Efficiency. The efficiency of the power supply shall be determined by comparing the input power to output power during rated input operation at room temperature, -55°C, and +71°C.

4.3.3.6 Overload. The power supply shall be subjected to the conditions of overload in 4.3.3.6.1 through 4.3.3.6.5. After completion of each test, with rated input, all electrical measurements shall comply with 3.4.1.

4.3.3.6.1 Submit 35 volts DC for one minute. Record output.

4.3.3.6.2 With rated input, subject the unit to five applications of a one second duration, arcing short circuit followed immediately by a one minute duration zero ohms short at the output. Record output.

4.3.3.6.3 With rated input, subject the power supply to a one hour zero ohm short circuit at the output. Record output.

4.3.3.6.4 With the rated input and output, subject the power supply input to five applications of +200 volts peak and five applications of -200 volts (peak). Each application shall be of one micro-second duration. Record output.

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4.3.3.6.5 With rated input and output, subject power supply input to five applications of +80 volts. Time duration of the 80 volt level should be at least 10 milliseconds. Record output.

4.3.3.7 Temperature - altitude tests. With unit inoperative, subject it to a temperature of -55°C and 70,000 feet pressure altitude for one hour. Operate unit at rated output for four hours, turning unit off once each hour. Then increase temperature to +71°C and 70,000 feet pressure altitude in not more than fifteen minutes and operate power supply at rated output for four hours. Turning unit off once each hour record input and output every 30 minutes of operation.

4.3.3.8 Vibration test. The power supply assembly shall be vibrated at an amplitude of 0.06 inches, total excursion, with the frequency varying uniformly from 10 to 55 to 10 cycles per second in approximately one minute. Output shall be monitored throughout test. Vibration shall be for ninety minutes in each of the following directions.

4.3.3.8.1 Apply test horizontally, parallel to the major axis of the unit.

4.3.3.8.2 Apply test horizontally, at right angles to the major axis of the unit.

4.3.3.8.3 Apply test vertically, at right angles to the major axis of the unit.

4.3.3.9 Environmental tests. The power supply shall be subjected to the following tests in accordance with MIL-T-5422.

a. Humidity See 3.4.5.2

b. Salt spray See 3.4.5.4

c. Fungus See 3.4.5.6

d. Explosion proof See 3.4.5.7

e. Shock See 3.4.5.8

4.4 Acceptance tests. MIL-E-5400 shall apply, except as noted. Acceptance test shall consist of:

a. Individual tests.

b. Sampling tests.

c. Reliability - life tests.

4.4.1 Individual tests. Each unit shall be tested according to 4.3.3.1 through 4.3.3.6.5 except 4.3.3.4 and 4.3.3.6.3. Each unit will be inspected visually to verify the requirements stated herein. When required additional inspections, as directed by the procuring agency, shall be made (see 6.2).

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4.4.2 Sampling tests. Two percent of each three month's production or three units, whichever is greater, shall be selected at random by the resident military inspector and subjected to the tests of 4.3.3 through 4.3.3.8.3. A test report including a detailed description of all failures, shall be forwarded to the procuring activity.

4.4.3 Reliability - life test. Three percent, or more, of each month's production shall be selected at random by the resident military inspector and operated at rated input and output to accumulate reliability - life test time. Acceptance or rejection of that month's production will be based on the following schedule:

Accumulated Hours	Failures Allowed for Acceptance	Number Failures for Rejection
600	2	3
1000	3	7
1500	4	10
2000	5	13
2500	6	15
3000	8	18
3500	10	20
4000	12	22
4500	14	24
5000	16	26
5500	18	28
6000	20	30

Example:

If 600 hours of total test time are accumulated with not more than one failure, an accept point is reached. If two failures have occurred, testing shall be continued to the next time level (1000 hours). If three failures occur during the 600 hours, a reject point is reached.

If an accept point is reached, that month's production is accepted and that month's testing may be terminated. If a reject point is reached, the month's production is rejected, delivery shall not be made, and the procuring agency shall be notified immediately, of the full details of testing, failures, and modifications proposed to eliminate the source of failure. Any modification will be incorporated at no cost to the Government. The modified test samples will then be returned to test and testing will be resumed from the zero time point.

Should the maximum time level be reached without a definite accept or reject point, the procuring agency will make a determination of acceptance or rejection based on the test and failure data.

4.5 Definition of failure. A failure shall have occurred any time the performance of the unit falls below the specified in 3.4, except that secondary causes, external to the unit, shall be excluded.

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4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Power supply units shall be preserved and packaged levels A or C in accordance with MIL-E-17555.

5.2 Packing. Packing shall be levels A, B or C in accordance with MIL-E-17555.

5.3 Marking. In addition to any marking required by the contract order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This unit is intended as a DC to DC power supply to replace the DA-184/ARN-14.

6.2 Conflict of requirements. In the event a conflict arises between this specification and other applicable documents, this specification shall govern.

6.3 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Level of packaging and packing desired (see 5.1 and 5.2).

c. Invitations for bids, contracts, and purchase orders should state the location of the preproduction test.

d. Additional inspections not included herein, if required (see 4.4.1).

Custodians:
Air Force - 82
Army - EL

Review Activities:
Air Force - 82
Army - EL

Preparing Activity:
Air Force - 82

Project No. 6125-0100

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity*)

SUBMITTED BY (*Printed or typed name and activity*) | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE

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